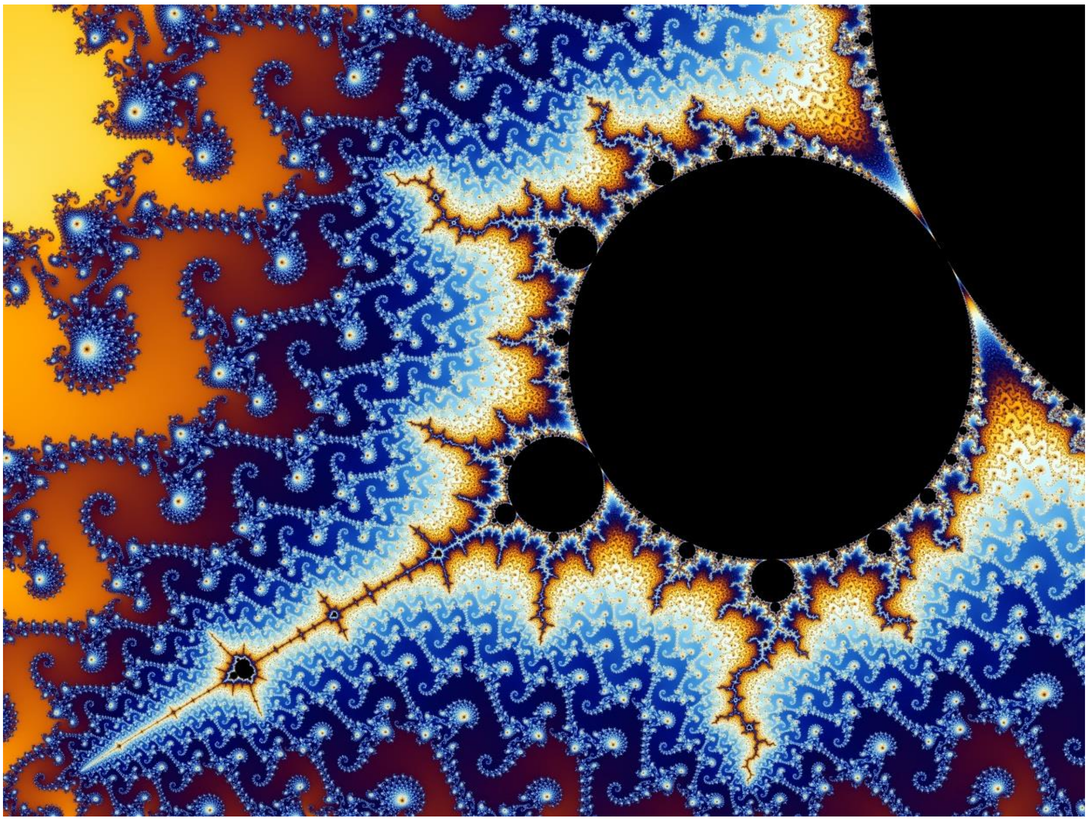
ST STITHIANS COLLEGE

Tour de Maths 2023

Leg 1

Question Paper - MEMO



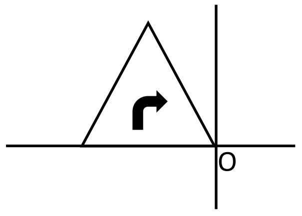
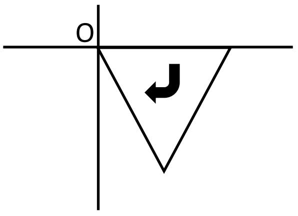
The Mandelbrot set

5 Mark Questions

Question 1	MEMO
If $p\otimes q$ means $3p+q^2$, then $(3\otimes 4)\otimes 5$ is equal to?	$(3\otimes 4)\otimes 5$ $=\left(3(3)+4^2\right)\otimes 5$ $=25\otimes 5$ $=3(25)+5^2$ $=100$ **ANSWER: 100**

Question 2	MEMO
How many factors does 2023 have?	The factors of 2023 are 1, 7, 17, 119, 289, 2023. Therefore, 2023 has 6 factors **ANSWER: 6**

Question 3	MEMO
Two integers a and b are defined as follows: $a=1+2+3+4+5+6+7+8+9+10$ $b=1\times 2\times 3\times 4\times 5\times 6\times 7\times 8\times 9\times 10$ What is the greatest common factor of a and b?	$a=1+2+3+4+5+6+7+8+9+10$ $a=55$ $a=5\times 11$ The Greatest common factor of a and b is 5 **ANSWER: 5**

Question 4	MEMO
The triangle shown is rotated 180° about O. Draw the outcome on your answer grid.  	 **ANSWER: as per diagram**

Question 5

Two dice are rolled. What is the probability that the sum of the two numbers shown is odd?

MEMO

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$P(sum\,is\,odd) = \frac{18}{36} = 50\%$$

ANSWER: $\frac{1}{2}$; $0,5$; 50%

Question 6

A bucket is filled half way with water. A cleaning liquid fills another 2 litres of liquid into the bucket. Now the bucket is $\frac{3}{4}$ full.

How many litres of fluid in total can fit into the bucket?

MEMO

Half full is 2 quarters to start.
Adding 2 litres adds a quarter to the bucket.
Therefore, the bucket must hold 4 x 2 litres = 8L

ANSWER: 8

Question 7

If $x > y$, solve for x and y in the equation: $x^3 + y^3 = 65$.

MEMO

$$x^3 + y^3 = 65$$

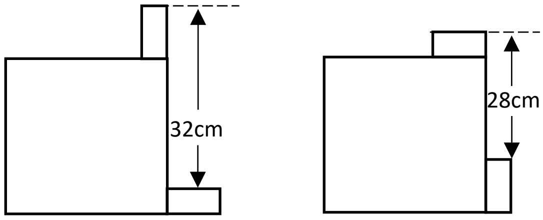
$$(x+y)(x^2 - xy + y^2) = 64 + 1$$

$$(x+y)(x^2 - xy + y^2) = (4+1)(16-4+1)$$

ANSWER: x=4 and y=1

Question 8

A large box and two identical small bricks are arranged in two ways as shown in the diagram below.



What is the height of the box?

MEMO

$$\frac{(32+28)}{2}=30cm$$

ANSWER: 30

Question 9

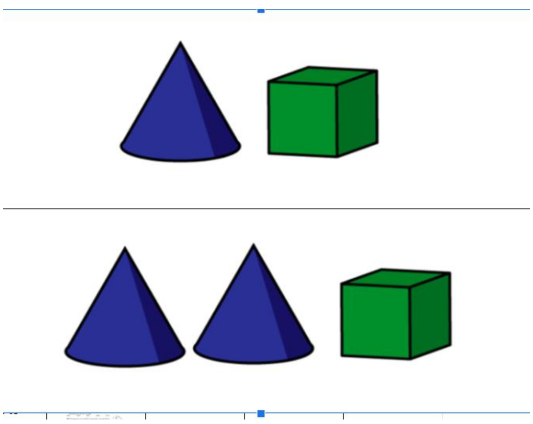
In recreational Mathematics, a square array of numbers, usually positive integers, is called a magic square if the sums of the numbers in each row, each column, and both main diagonals are the same. Using the numbers 1 to 9, complete the magic square below using the given information and transfer your answer to the answer sheet.

6	1	8
7	5	3
2	9	4

Question 10

The combined mass of the 2 shapes in the top image is 56 g. The combined mass of the 3 shapes in the bottom image is 72 g. Cones and cubes are identical throughout.

What is the mass of a cube?



MEMO

$$(\Delta+\square)=56$$
$$\Delta+(\Delta+\square)=72$$
$$\Delta+56=72$$
$$\Delta=16$$
$$(16+\square)=56$$
$$\square=40$$

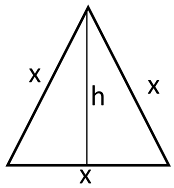
ANSWER: 40

10 Mark Questions

Question 11

It is given that an equilateral triangle has a perimeter numerically equal to its area. What is the side length of triangle?

MEMO



$$h=\sqrt{x^2-\left(\frac{1}{2}x\right)^2}$$
$$=\sqrt{\frac{3}{4}x^2}$$
$$=\frac{\sqrt{3}x}{2}$$

$$\text{Perimeter} = \text{Area}$$
$$3x=\frac{1}{2}x.\frac{\sqrt{3}}{2}x$$
$$12x=\sqrt{3}x^2$$
$$0=\sqrt{3}x^2-12x$$
$$0=x\left(\sqrt{3}x-12\right)$$
$$x=0 \quad or \quad x=\frac{12}{\sqrt{3}}$$
$$n.v$$

ANSWER: $\frac{12}{\sqrt{3}}=4\sqrt{3}\approx 6,93$

Question 12

What is the sum of the digits when we multiply 2^{2023} by 5^{2020} ?

MEMO

$$2^{2023}\times 5^{2020}$$
$$=2^3\times 2^{2020}\times 5^{2020}$$
$$=2^3\times\left(2\times 5\right)^{2020}$$
$$=8\times 10^{2020}$$

Therefore 8 with a whole lot of zeroes

Sum of digits is therefore 8 + zeroes

ANSWER: 8

Question 13	**MEMO**
Let a,b,c and d be four numbers with $a<b$ and $c<d$. The average of a and b is c. The average of c and d is b. If $d-a=60$, what is the value of $b-c$?	Ordering these numbers based on the information we get the following: a ; c ; b ; d If distance, $d-a=60$ then the symmetry of the averages means there are 20 units between each pair. Therefore $b-c=20$ **ANSWER: 20**

Question 14	**MEMO**
Lots of different positive whole numbers were written on a blackboard. Exactly 2 of these numbers are divisible by 2, and exactly 13 of these numbers are divisible by 13. The biggest number on the board is "*m*". What is the smallest value that "*m*" can be?	Exactly 2 of these numbers are divisible by 2 means we have two even numbers on the board. Exactly 13 of these numbers are divisible by 13, meaning some have factors 13 and/or 2. Our set then must contact 2 even numbers and the rest are odd multiples of 13. $Set=\{13;26;39;52;65;91;117;143;169;195;221;247;273\}$ for these criteria to be me the smallest value of m is 273 **ANSWER: 273**

Question 15

On a pond 16 lilly pads are arranged in a 4x4 grid as shown in the diagram below. A frog sits on a lilly pad in one of the corners of the grid (as shown on the picture). The frog jumps from one lilly pad to another horizontally or vertically. In doing so he always jumps over at least one lilly pad. He never lands on the same lilly pad twice.

What is the maximum number of lilly pads, including the one he is sitting on, on which he can land?

MEMO

1	9	2	
5	7	4	6
12	10	13	11
Start	8	3	

1	9	2	10
5	7	4	6
12		13	11
Start	8	3	

ANSWER: 14

20 Mark Questions

Question 16

What is the remainder when 2^{645} is divided by 7 ?

MEMO

$\operatorname{Re} m\left(\frac{2^1}{7}\right)=2$ $\quad$ $\operatorname{Re} m\left(\frac{2^2}{7}\right)=4$ $\quad$ $\operatorname{Re} m\left(\frac{2^3}{7}\right)=1$

$\operatorname{Re} m\left(\frac{2^4}{7}\right)=2$

We notice that the remainder cycles 2, 4, and 1.

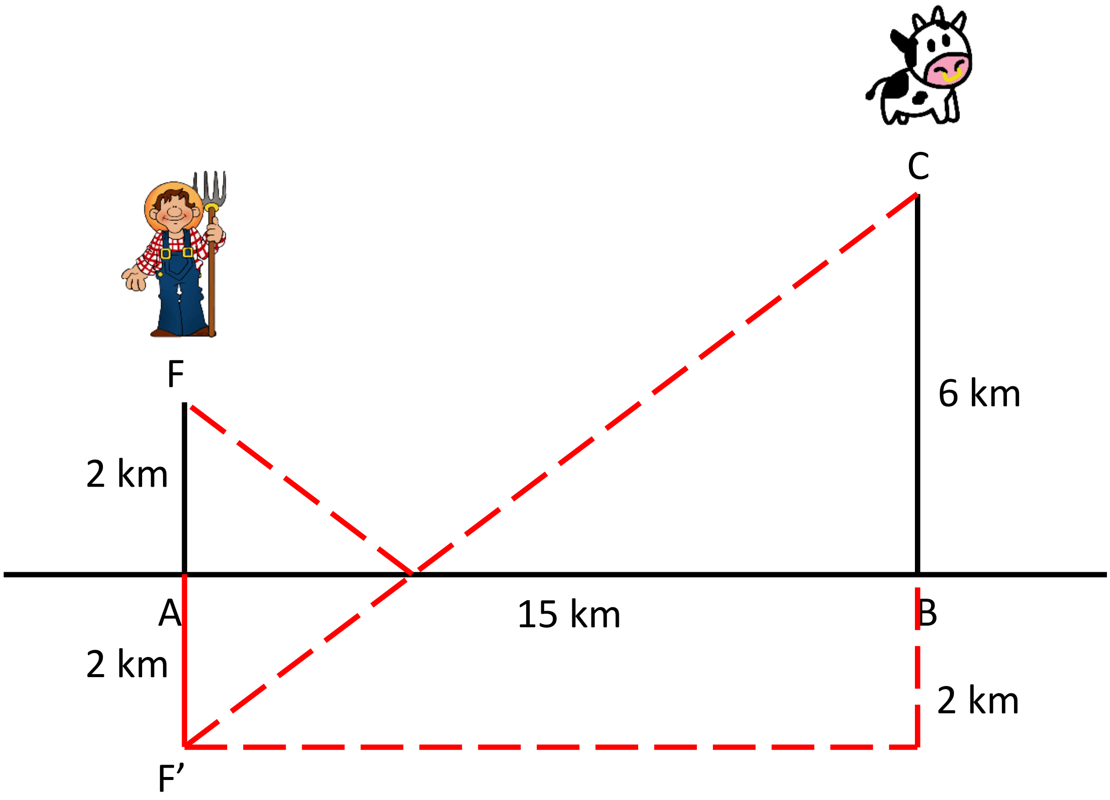
$$\frac{645}{3}=215 cycles$$

ANSWER: 1

Question 17

A farmer (F) is 2km away from point A. A and B are 15 km apart and represent a stretch of river on the farm. A cow (C) is 6km from point B. The farmer wants to take water to the cow who cannot move. The farmer will walk to a point between A and B and then walk with the water to the cow. The farmer wants to minimize the distance he wants to walk in total.

What is the minimum distance the farmer walks to collect water and take it to the cow?

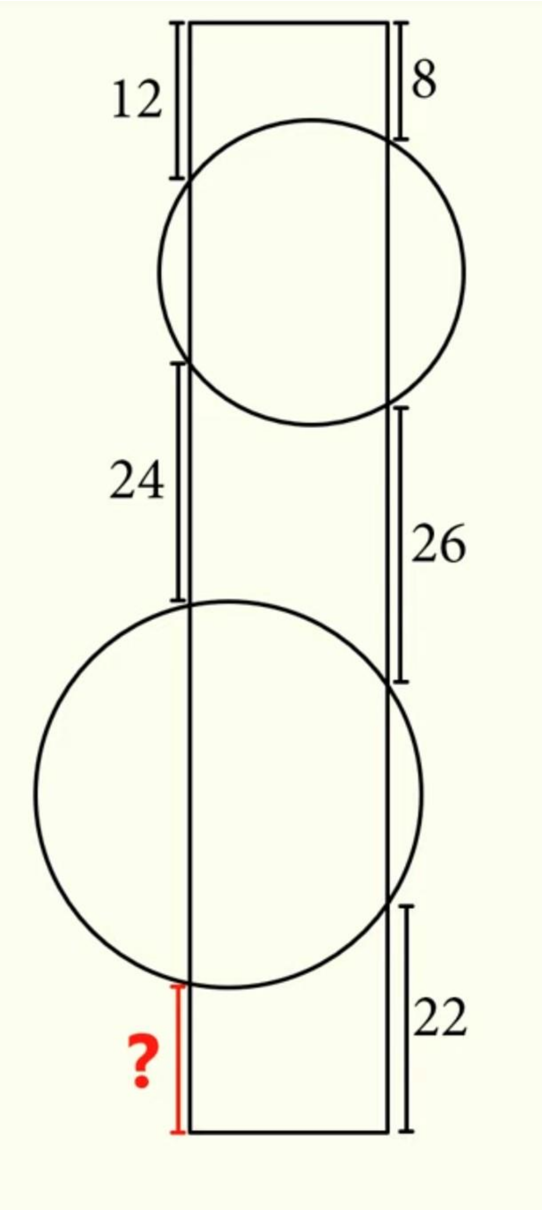
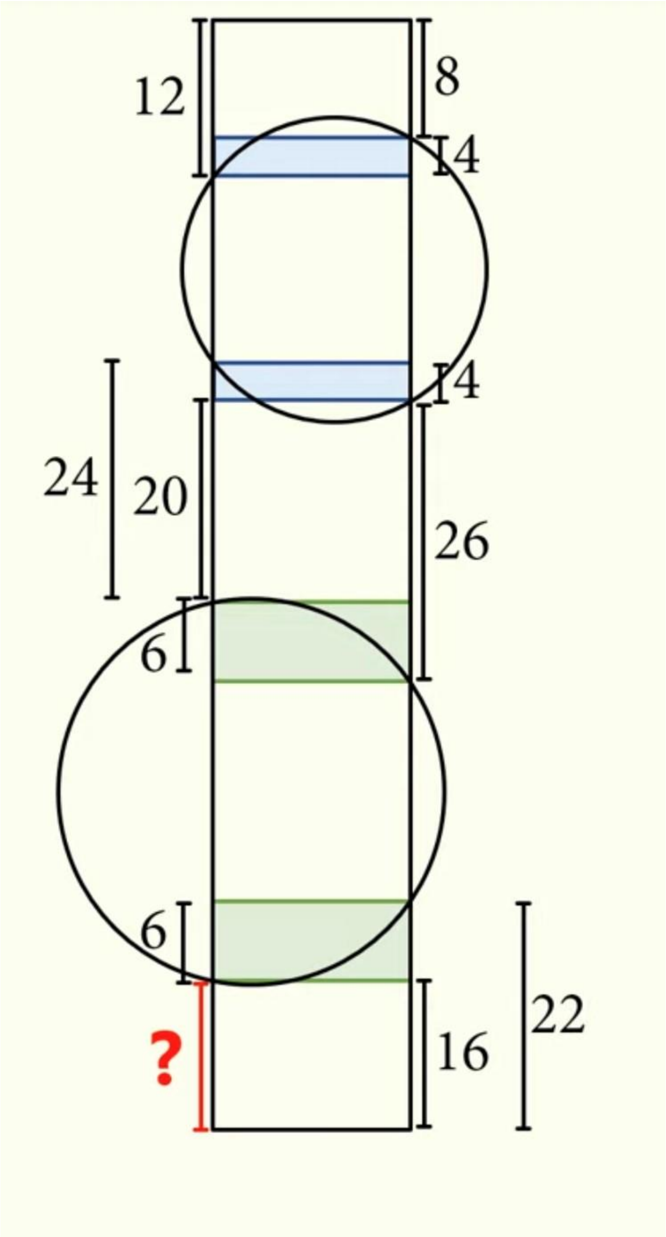


$$F'C=\sqrt{(6+2)^2+15^2}$$
$$=17km$$

ANSWER: 17

Question 18	MEMO
In the equation $N \times U \times (M + B + E + R) = 33$ each letter represents a different digit $(0,1,2,...,9)$ In how many different ways can the letters be replaced by different digits?	$N \times U \times (M + B + E + R) = 33$ $N \times U \times (M + B + E + R) = 3 \times 11$ $N \times U \times (M + B + E + R) = 1 \times 3 \times 11$ $N \times U \times (M + B + E + R) = 1 \times 3 \times (0 + 2 + 4 + 5)$ The order of elements Infront of the brackets: $2!$ The order of elements in the bracket can be described by $4!$ Total order of elements is $2! \times 4! = 48$ **ANSWER: 48**

Question 19	MEMO
A bag contains 5 sticks of length 2; 3; 4; 5; and 6 units. How many different triangles can be formed from these sticks?	The sides of a triangle rule says that the sum of the lengths of any two sides of a triangle must be greater than the third side. The possible combinations of this are: $\{2,3,4\}$ $\{2,4,5\}$ $\{3,4,5\}$ $\{3,4,6\}$ $\{3,5,6\}$ $\{4,5,6\}$ **ANSWER: 6**

Question 20	MEMO
 	 **ANSWER: 16**

Total: 200

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